DECLAN TRAINOR

Huntington, NY | (516) 660-9272 | tdeclan58@gmail.com | linkedin.com/in/declan-trainor-385a29371
U.S. Citizen | Eligible for U.S. Security Clearance

EDUCATION

The Pennsylvania State University — **University Park, PA**
College of Engineering — *Expected: May 2028*
Bachelor of Science in Electrical Engineering — **GPA: 3.30/4.00**
Relevant Coursework: Continuous-Time Linear Systems; Electronic Circuit Design 1; Design Tools; Circuits and Devices

PROFESSIONAL EXPERIENCE

Rodelco Electronics — **Ronkonkoma, NY**
Engineering Intern, Defense & Avionics Hardware — *May 2026 – Present*

- Develop VHDL firmware on an Intel FPGA to control an AD9914 direct digital synthesizer over SPI — initialization sequencing, DAC calibration timing, and digital ramp generator configuration — producing linear frequency Ramp chirp, frequency-hopping, and Barker-coded phase modulation waveforms across a 400–500 MHz band
- Isolated a control-register defect in simulation and on hardware: a stray programmable-modulus enable bit in CFR2 was locking the part to a fixed output instead of sweeping; validated the corrected register map against an AD9914 evaluation board
- Converted legacy schematic-based FPGA designs into maintainable VHDL RTL, including an eight-ROM waveform generator using a multiplexer to select among eight custom MIF-initialized waveforms, verified with self-checking ModelSim testbenches before committing to hardware
- Characterized low-pass, high-pass, and band-pass filters on a Keysight PNA-L vector network analyzer ($140k asset), performing full 2-port calibration and S-parameter measurements, and analyzed PSG signal generator phase-noise data with noise-floor correction
- Developed a control board test procedure now used team-wide to verify operating frequencies and supply voltages, including the safety steps for powered testing
- Ran temperature and data characterization tests on a coupler detector using a network analyzer and Keysight VEE Pro

Eastern Fence LLC — **Medford, NY**
Distribution Associate (Seasonal) — *May 2025 – Aug 2025*

- Operated $25k forklifts and coordinated order fulfillment for loads valued from $500–$10k in a high-volume distribution center, balancing throughput with accuracy to meet contractor deadlines

TECHNICAL PROJECTS

Audio Tone Control & Karaoke Processor — **University Park, PA**
EE 210 Design Project — *Spring 2026*

- Designed a five-stage analog signal chain around LF412 op-amps: an inverting summer/difference amplifier with selectable mixer and vocal-cancellation modes, a Baxandall bass/treble tone control, volume control, a four-LED comparator volume display, and an output buffer
- Hand-calculated component values, verified in Multisim, prototyped on a breadboard, then soldered and demonstrated a working PCB on a ±9 V supply

LEADERSHIP & CAMPUS INVOLVEMENT

Penn State Men's Rugby — **University Park, PA**
Division I Athlete- Top 15 team in the country — *2024 – 2025*

- Balanced a Division I practice and competition schedule with a full engineering course load while maintaining a 3.30 GPA

IEEE, Penn State Chapter — Member — *University Park, PA | 2024 – Present*
Phi Kappa Tau Fraternity — Member — *University Park, PA | 2024 – Present*

SKILLS & INTERESTS

Languages: VHDL, Python
FPGA & EDA Tools: Intel Quartus Prime, ModelSim, RTL simulation, self-checking testbenches, MIF/ROM initialization, JTAG programming, timing and pin assignment
Test Software: LabVIEW; Keysight VEE Pro, Multisim, SolidWorks, Excel data analysis
RF & Test Equipment: Vector network analyzers (Keysight PNA-L), S-parameter measurement and 2-port calibration, PSG signal generators, spectrum and phase-noise analysis, oscilloscopes, DMMs
Engineering: Direct digital synthesis (DDS), SPI and register-level device control, finite state machine design, test-procedure development, schematic interpretation, PCB soldering
Training: Google AI Essentials- prompt engineering/vibe coding; Forklift Training; Sailing Instructor; CPR; Lifeguard